

The DIY Hoverbike — Lifting 150 kg with Four Cymbals and a \$12 Microcontroller.pdf

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Summary

This technical guide details the construction of a **150 kg single-rider hovercraft** utilizing the principles of **near-field acoustic levitation (NFAL)**. By repurposing **resonant bronze cymbals** as acoustic radiators and driving them with a low-cost **microcontroller**, the authors demonstrate how to achieve stable flight in ground effect using **conventional physics**. The document serves as both a **practical DIY manual** for a budget-friendly vehicle and a foundational prototype for more advanced research into **dark-sector propulsion** and space-capable architectures. Key design innovations include a **broadband multichord drive** for energy efficiency and a **unified passive safety system** consisting of a parachute-shaped inflatable for impact protection.

Keywords: Near-field acoustic levitation, DIY hoverbike construction, Multichord drive electronics, Ground effect physics, Unified safety system

Content

[\[Image\]](#)

Companion in the Boundary Problem paper network. Root: The Shape of the Universe. The DIY Hoverbike **Lifting 150 kg with four cymbals, twelve piezos, and a \$12 microcontroller. Osika & collaborators, draft 0.3, 2026-05-22. Figure 1. Production-intent rendering of the four-cymbal hoverbike. Four 20" bronze cymbals, mounted bell-down at the corners, generate a near-field acoustic pressure floor that supports the bike in ground effect. The body is a fairing over the topology-optimised frame; the rider sits saddle-style with the handlebar providing both directional steering (yaw, pitch,**

roll, lift) and the spring-loaded 6-DoF throttle/control inputs described in §3.7. Acoustic emission is confined to a narrow downward cone by the cymbal geometry; what's visible above the cymbals is structural and electronic enclosure only. A 150 kg single-rider hover vehicle built from four 20" bronze cymbals (or stainless mixing bowls, in the budget variant), each with three piezo transducers drilled through it, mounted bell-down at ground effect on ~50-100 W using nothing but classical near-field acoustic levitation. Differential thrust between the four cymbals handles steering, quadrotor-style. No new physics required. Build time: one weekend.

Companion papers. This document covers the buildable atmospheric vehicle and the conventional physics behind it. For the experimental program measuring the $\chi\nu$ susceptibility — the hypothesised coupling between coherent matter and the dark-sector interpolation function — see **Probing the OPH Dark Sector via Coherent-Matter Susceptibility (Osika & Mueller 2026a)**. For the extended architecture across the full altitude range including space propulsion **and the Tic-Tac-class performance envelope, see How to Build a UFO — the Saucer Conduration (Osika 2026c)**. The hoverbike is the ground-effect endpoint of that broader architecture; the ~500 build described here is the prototype anyone can construct in a weekend.

§1 What this is and isn't This paper covers exactly one thing: how to build a 150 kg single-rider hover vehicle that works in atmosphere, in ground effect, using nothing but conventional acoustic physics. No new physics is required for the hoverbike itself to function. The underlying mechanism — near-field acoustic levitation (NFAL) through a multichord-driven array of resonant metal bowls — has been demonstrated in the laboratory for forty years with small loads. We are scaling it up. This paper does not cover: The Mueller-Osika $\chi\nu$ susceptibility framework, except as a brief reference to a hypothesis being measured experimentally on the same hardware. See companion paper 1 for that physics. The saucer-class architecture that operates outside ground effect, in space, or with non-trivial $\chi\nu$ coupling. See companion paper 3 for that. Any of the high-amplitude regimes that produce the Tic-Tac-class sensory effects historically reported for UAP. Also paper 3. The hoverbike works at $\chi\nu = 0$. If $\chi\nu$ turns out to be non-zero, the vehicle gains additional capabilities (vacuum operation, altitude independence) but the atmospheric hover described here is conventional physics.

§2 The physics, briefly substrate (the vehicle's bottom) and a hard reflector (the ground). The standing wave exerts a time-averaged radiation pressure on the substrate (and an equal-and-opposite pressure on the ground), lifting the substrate upward. The cavity Q (which determines how many cycles of acoustic energy circulate per cycle of dissipation) is what makes the lift efficient: most of the acoustic energy stays in the cavity rather than being radiated to far field. Two design choices push NFAL into the human-carrier regime: 1. Resonant metal substrates. Each cymbal is an engineered acoustic radiator with a rich eigenmode spectrum. A bronze cymbal of the kind used

in drumming has been tuned by centuries of practical work to produce efficient acoustic radiation across multiple frequencies. The cymbal's bell acts as a focused central source; the bow as a broader distributed source. Mounted bell-down, the entire shape funnels acoustic energy toward the ground.

2. Broadband multichord drive. Twelve transducers driven by coordinated phase-locked signals at a spectrum of frequencies tuned to the cymbal eigenmodes. The total stored coherent vibrational energy across the substrate-plus-air system scales with the number of modes excited, and the per-mode Stokes boundary-layer loss is lower than what you'd get from a single-frequency drive at the same lift.

2.1 The substrate-plus-air coherent state When a substrate vibrates and creates a standing wave in the surrounding air, both the **substrate and the air column it drives are in coherent states. For NFAL in atmosphere, the air** cavity stores far more coherent energy than the substrate itself — for typical operating conditions, ~20× more. The substrate's job is mostly to establish and maintain the air's coherent state. This matters for the design: we don't need exotic high-Q crystal substrates because the air is where most of the action is. A thin metal bowl (or even thin plastic) works fine, as long as it radiates efficiently into the air cavity.

2.2 The Stokes floor for atmospheric NFAL The irreducible loss in NFAL is the air's viscous boundary layer at the no-slip walls (substrate above, ground below). Per unit area: $P_{\text{Stokes}}/A = \frac{1}{4} \omega^2 \cdot \sqrt{(2\mu\rho\omega)}$ **For fixed lift force per area ($F/A = \rho u^2$), this simplifies to:** $P_{\text{Stokes}}/A = (F/A) \cdot \sqrt{(\mu\omega / (2\rho))}$ So lower ω means lower Stokes loss per unit lift. The optimum sits in the 200 Hz to 2 kHz band for sensible cavity geometries (the wavelength has to fit in the vehicle-to-ground gap; the substrate displacement has to be physically realisable). For four 20" cymbals (total area ~1 m²) lifting 150 kg at 340 Hz center frequency: $P_{\text{Stokes}} \approx 90$ W This is the floor — what you'd need with perfect electronics, perfect substrate efficiency, and no other losses. Realistic operation is in the 100–500 W range depending on ground type (concrete absorbs more than polished surfaces), altitude (further from ground reduces cavity Q), and how aggressively the drive is optimised.

Doctrine sentence

- **The DIY hoverbike works on conventional NFAL physics, scaled up via four resonant bronze cymbals and broadband multichord drive. Atmospheric hover floor: ~90 W in ideal conditions. Realistic hover: ~100-300 W depending on operating conditions. No new physics required for the vehicle itself to function.**

§3 The build

3.1 Substrate — four bronze cymbals The default substrate is four 20" (510 mm) bronze ride/crash cymbals, sourced from a music store or the used market. Sabian or Zildjian are good options. Total cost for a used set: ~£200. The cymbals are engineered for efficient acoustic radiation of complex spectra — literally their designed purpose — which is also what we want. Alternative materials, in descending order of merit: Material Source Thickness Cost Notes 20" bronze cymbals Music store, used 0.5–1.5 mm £30– 100 each Default. Best radiation efficiency and aesthetics. Drum gongs (16–24") Music store 1–3 mm

□50– 200 each Heavier; better if you want eventual vacuum capability. Material Source Thickness Cost Notes Stainless mixing bowls IKEA, restaurant supply 0.4–0.8 mm □5–15 each Cheapest. Whole vehicle for < □400. Use for 88rst prototype. Spun aluminium lampshades Lighting suppliers 0.5–1 mm □15–30 each Lightest. Acoustic properties OK. 3.2 Geometry & mounting Diameter: 400–510 mm per cymbal. Four cymbals give ~0.5–0.8 m² of total substrate area. **Orientation: bell pointing downward. The small raised dome at the cymbal's centre faces** the ground. This is upside-down relative to a normal drum-kit mounting. Bell-down does two useful things: the bell becomes a small focused acoustic source aimed at the ground along the central axis, and the bow (the curved outer surface, now facing downward) acts as a shallow parabolic re88flector funnelling acoustic energy downward. **The cymbal's natural shape is the centrally-peaked ð eld shape the NFAL standing wave** wants. Outward angle: tilt each cymbal 10–15° outward from vertical, in a square pattern. At zero differential drive this produces pure vertical lift (the four horizontal components cancel by 4-fold symmetry); under differential drive, the imbalance produces horizontal thrust. Same principle as a quadrotor drone — see §3.6. Transducer placement: three holes per cymbal, equispaced at 120°, at ~60% of the radius from the centre. This avoids the rim (high strain, structurally weakest after drilling) and the dead centre (high strain for $m=0$ modes). Use a step drill with a centre punch 88rst — thin metal deforms if drilled with a regular twist bit. Transducer mounting: M3 or M4 bolts through the drilled holes, active face on the concave (top) side of the cymbal, wiring on the convex (bottom) side. Use rubber gaskets for acoustic sealing around the bolts. 3.3 Transducers 12× 60 W piezo disc transducers, ~30 mm diameter (3 per cymbal). Driven independently by 12 class-D amplifier channels. Mouser/AliExpress part numbers omitted; the right ones change by Tuesday. Look for “Langevin” or “sandwich” piezo transducers in the kHz range with M-bolt mounting. 3.4 Driver electronics Waveshare or Pimoroni dev board with a 2” LCD, 6-axis IMU, microSD, battery monitoring, and a rotary encoder (for throttle) all on-board. The RP2350's PIO blocks generate 12 phase-locked PWM channels at the broadband multichord spectrum; ADC channels read back a pickup piezo for resonance locking and the IMU for altitude and tilt; the remaining compute runs the multichord coordinator and the warm-loop tuner. The rider sees altitude, lift balance, and the kill-switch state on the LCD. Firmware is a direct port of the 12_PORT_SUBSTRATE_ADAPTER from the laser-toroid optical chambers (Mueller & Osika 2026 SHA-256d paper). Above the adapter — multichord coordinator, phase scorebook, resonance-locking warm loop — the code transfers unchanged. Below the adapter, the per-port output stage swaps from laser current drive to class-D PWM into a piezo. The 12 channels split as 3 per cymbal × 4 cymbals; the coordinator handles inter-cymbal phasing as well as intra-cymbal mode selection. 3.5 Frame Aluminum tube frame holding the four cymbals in a square pattern ~600–800 mm on a side, with the rider seated above the centre of

the pattern. Mass: ~3 kg for a minimal welded or fastener-assembled frame using T-slot connectors and standard motorbike-style ergonomics (saddle, footpegs, handlebar). No 3D printing required, though a topology-optimised PETG print works too and shaves ~1 kg.

3.6 Steering via differential thrust

Each cymbal is independently controllable in amplitude by the multichord coordinator. At identical drive across all four, the 10–15° outward tilts produce horizontal force components that cancel by 4-fold symmetry, giving pure vertical lift. Modulating per-cymbal amplitudes asymmetrically produces a net horizontal thrust vector: Forward / back / sideways: increase opposite-side cymbal, decrease same-side cymbal. Yaw: increase opposite-diagonal pair, decrease the other pair. Roll / pitch: same modulation pattern as forward/back; feels different to the rider (rotational vs translational). No separate steering thrusters. The same 12 channels that drive lift also drive steering, via software amplitude modulation per cymbal. The rider's control inputs (throttle, joystick) translate to per-cymbal amplitude scalings inside the RP2350 firmware.

3.7 Battery and control system

motorcycle's controls. The cleanest expression has two components: one input device, one output device.

Battery: four hot-swappable e-bike-class packs Four 1.5 kWh packs (~5 kg each, ~£300 each in Bosch PowerTube format) slot into the central spine where a motorcycle's fuel tank would be. 6 kWh total, ~20 kg. Each pack quick-releases in about 10 seconds; carry spares in cargo for hot-swap during long rides. At hover power (~100 W), one pack lasts ~15 hours; at sport cruise, ~3 hours; in heavy sport-mode use, ~30-60 minutes.

Control: one spring-loaded 6DoF handlebar The entire handlebar is mounted on a precision-damped 6DoF gimbal with ~1-2 cm of travel against centering springs. The rider grips the bar normally; the bar itself moves a small amount in response to applied force. Push forward / pull back → horizontal acceleration Push left / right → lateral translation Push down / pull up → descent / ascent Twist → yaw Tilt forward/back → pitch (nose up/down) Tilt left/right → roll Release the bar and it springs to center; the vehicle decelerates to a stationary hover. Grip with both hands (capacitive sensors) to wake the vehicle from idle. Press the single red button on top of the right grip for a controlled emergency descent. No twist grips. No mode selector. No foot pegs. No throttle. The bar's spring resistance gives the rider direct tactile feedback for the magnitude of commanded acceleration. A force sensor in the grip quietly caps commanded acceleration at what the rider's current grip strength can physically hold — if the rider loses grip while accelerating hard (about to fall off the back), the cap drops and the vehicle eases up. The human-grip-strength is in the control loop; the bike physically cannot throw the rider off. Display: helmet visor HUD Altitude, speed, battery level. Projected via Bluetooth onto the rider's helmet visor as a thin overlay. The handlebar itself has no display. The rider is looking forward; the information goes where the rider is already looking. Walk up to the bike. Grip the handlebar with both hands — the bike wakes, lifts off, hovers at 30 cm. Push the bar in whatever direction you want to go; the harder you push, the stronger the

response. Release the bar at any time and the vehicle stops and holds position. Press the red button to descend and land. The whole interface is the bar and the visor HUD. There are no modes, no menus, no settings. The vehicle has one response curve, tuned to be safe at maximum input. Beginners push gently because they don't yet know how the bike responds; expert riders push hard. Same vehicle, same physics; what changes is the rider's commitment. **Doctrine sentence · One input device. One emergency override. One presence sensor. One output channel. The hoverbike's control system has fewer components than a kettle. Everything that's here is load-bearing; everything else got removed.**

3.8 Performance envelope With 6 kWh of available battery, ~40 kW of peak transducer power (12 piezos at 5× their continuous rating, briefly), and the air-cavity acoustic limit at Mach 1 particle velocity, the performance envelope is set by the rider's tolerance for acceleration, not by the vehicle's capability. Maneuver Acceleration Power (peak) Duration Hover 0 g ~100 W hours Gentle climb 1 g ~250 W hours Sport launch 3-5 g ~1-2 kW tens of seconds Fighter-jet maneuver 9 g ~3 kW seconds Emergency dodge (clamped) 15 g ~6 kW ~1 second For comparison: a Ducati Panigale V4 does 0-100 km/h horizontally in ~3 seconds at 1 g. **The hoverbike does 0-100 km/h vertically in ~0.6 seconds at 5 g sport-launch.** Top speed is bounded by acoustic-cavity stability in lateral motion at ~250 km/h. The actual ceiling on rider experience is human consciousness — the vehicle can do more than the rider can survive. The grip-strength acceleration limit (above) means the rider cannot accidentally command an acceleration they can't physically hold against. The kill switch lands the vehicle in ~3 kill drive power produce 1 g freefall; from typical 30 cm hover height, this is a survivable drop onto the bike's frame.

3.9 Safety equipment One unified passive safety system handles every failure mode the vehicle can encounter: a parachute-shaped inflatable deployed from a CO2 cartridge in the central spine. When inflated, it forms a 6 m bright orange disc above the vehicle, with the vehicle suspended beneath. The same physics that gives a parachute its descent rate gives the airbag its drag; the same fabric structure that supports the canopy cushions impact; the same enclosed volume that contains the inflation gas provides buoyancy on water. Geometry and physics 6 m diameter parachute-shaped inflatable (large central disc with a vent for stability). When inflated: Frontal area: 28 m², drag coefficient $C_d \approx 1.4$ (parachute shape). Terminal velocity in air: ~7.7 m/s — equivalent to a 3 m drop onto pavement. Same descent rate as a Cirrus SR-22 BRS parachute. Inflated volume: ~17 m³, giving ~17,000 kg of buoyancy on water — over 100× safety factor for vehicle + rider. Impact cushioning: structure progressively debounces under landing load, dissipating energy across ~1 m of structural compression. Peak rider deceleration capped near 8-10 g for a controlled stop. Visibility: bright orange (or whatever colour you like for the "ejection" aesthetic); visible from miles away for search-and-rescue. Deployment A CO2 cartridge in the central spine fires when the flight computer detects an impact

(acceleration $>5\text{ g}$), excessive descent rate (vertical velocity $>5\text{ m/s}$ sustained), or manual button activation. Inflation completes in ~ 2 seconds. Above $\sim 20\text{ m}$ altitude: full inflation completes before impact. Terminal velocity reached, gentle landing. Below $\sim 20\text{ m}$ altitude: partial inflation at impact, but the cushioning is sufficient because the fall distance is short. Direct fall from 10 m onto an inflating airbag has very different injury physics from the same fall onto pavement. PPE, and the partial inflation collectively reduce impact severity to typical motorcycle-low-speed-crash levels (survivable). Mass and cost Mass: $\sim 15\text{ kg}$ (canister + fabric + plumbing + control electronics). Cost: $\sim \$2000\text{--}4000$ for marine-inflatable-grade materials and pyrotechnic deployment. Rebill: $\sim \$100$ to recharge the CO_2 cartridge and repack the fabric after a non-event activation. This single system replaces what light aircraft typically split across two: the ballistic parachute (for descent rate) and the cabin crash protection (for impact absorption). At our scale and payload, a parachute-shaped inflatable does both with one component, with the added bonus of water-landing rotation. Rider personal protective equipment — helmet, gloves, protective jacket, boots, and integrated harness anchor to the vehicle frame — is required for all operations. $\sim 3\text{ kg}$ of PPE per rider. **Doctrine sentence · One safety system. Six metre inflated disc, deploys in two seconds, slows the vehicle to parachute-class descent rate, cushions the impact, floats on water, visible from miles away. It is what you would design if you were starting from scratch and refusing to inherit the “parachute plus airbag” architecture of legacy aviation.**

3.10 Mass budget Component Mass Notes Rider 80 kg $4\times$ bronze cymbals ($20''$) $4\text{--}6\text{ kg}$ Up to $\sim 25\text{ kg}$ if you choose drum gongs. $12\times 60\text{ W}$ piezo transducers 1.2 kg Aluminum tube frame 3 kg Topology-optimised PETG variant: 2 kg . Battery ($4\times 1.5\text{ kWh}$ hot-swap packs, 6 kWh total) 20 kg Hot-swappable. $\sim 3\text{ hr}$ sport cruise, hours of gentle hover. Electronics (RP2350 + class-D channels + wiring) 0.4 kg Whole flight computer fits in a palm. Component Mass Notes Rider PPE (helmet, gloves, jacket, boots, harness) 3 kg Required. Unified passive safety system (parachute-airbag) 15 kg Standard equipment, $\$3.9$. Covers all altitudes and water landings. Total $\sim 127\text{ kg}$ 23 kg payload margin under 150 kg design point. Cheapest build (IKEA bowls + commodity piezos + aluminum tube): under $\$400$ total bill of materials. Mid-tier build (used bronze cymbals + quality piezos): $\sim \$800$. No expensive variant unless you go to gongs for the heavy-substrate option. $\$4$

The first-light experiment Before trying anything, run the pot-lid first-light experiment to characterise the multichord drive and verify the substrate adapter is doing the right thing.

4.1 Pot-lid setup A 200 mm aluminum pot lid ($\sim \$15$ from any kitchen store), $\sim 0.5\text{ mm}$ thick, with 12 piezos arranged around the perimeter in a ring pattern. Hung from a 2 m USB cable acting as a pendulum from a ceiling joist. The piezos drive the lid at its eigenmodes; horizontal deflection of the pendulum under any xv force (or just acoustic radiation pressure) is measured by a laser interferometer or simple LED-photodiode setup pointed at the lid. This is also the apparatus described

in detail in companion paper 1, where it's used as the primary χ_v measurement. For our purposes here, the pot-lid is a calibration rig: it confirms that the multichord drive produces the expected acoustic radiation pressure at measurable amplitudes and that the substrate adapter firmware generates clean phase patterns.

4.2 What you should see

At conservative drive amplitudes (~ 10 W per piezo, well below yield), the pot lid should deflect under acoustic radiation pressure by a few microns to a few millimetres depending on drive frequency and resonance proximity. This is conventional acoustic physics; no new effects are required. The point of the experiment is to:

1. Verify the firmware generates clean phase-locked PWM
2. Verify the piezos drive the substrate at its eigenmodes
3. Identify any artefacts (thermal expansion, electrostatic, EMI) that could mimic χ_v signals in the more sensitive measurements of companion paper 1

If the pot lid behaves as expected, the firmware and electronics are validated and you can proceed to building the full hoverbike. If it doesn't, debug the drive chain before scaling up.

§5 Things that will go wrong

Field guide compiled from our own builds:

1. The cymbals will deform on first overdrive. Bronze yields at lower stress than you'd **guess. Ramp the drive amplitude slowly. Have spare cymbals.**
2. The piezos will desolder above 60°C . Heatsink them. A 1 mm copper foil bonded to the bottom of each piezo and ducted to the chassis is enough. Lead-free solder helps.
3. The neighbours will object. Even at sub-audible fundamental frequencies (~ 1 kHz), the harmonics and the structural vibrations of nearby objects are loud. Build in a garage or rural location.
4. The first lift will be unstable. The substrate-plus-air coherent state is sensitive to standoff distance; cavity properties change by factors over a few centimetres of altitude. The control loop has to follow altitude with sub-centimetre response time. Use the IMU accelerometers, not ultrasonic rangefinders (your own drive saturates the latter). Expect to spend a week on the height-stabilisation PID.
5. You will accidentally levitate something you did not mean to. Dust, small electronics, light pieces of paper. This is harmless and, frankly, the best part.
6. The first flight will not have a smooth power curve. The cymbals have eigenmodes at specific frequencies; the multichord coordinator has to track them as they shift with temperature, drive amplitude, and bolt torque. The warm-loop tuner handles this in steady state but lags during transients.
7. Ground type matters a lot. Polished concrete: great. Carpet: terrible (absorbs the acoustic energy). Lawn: bad. Sand: worst. The cavity Q depends on the ground's acoustic reflectivity. Plan first-flight venue accordingly.

§6 Safety

This is real safety, not jokes. High-amplitude acoustic drive can damage hearing via subharmonics and the drive is on. The acoustic field can heat soft tissue. At the SPL levels needed for 150 kg lift, exposure times of seconds can cause localised burns. Do not put your hand or face between the substrate and the floor while running. The substrate can resonate at frequencies other than the design band. A safety stop button that cuts power on a 100 ms timer should be within arm's reach at all times. Tether the early prototypes. A simple steel cable from a strong

overhead anchor, with enough slack for the design flight envelope, prevents the worst failure class (drive cuts out at altitude with a 130 kg falling object containing a person). Do not test untethered until at least ten consecutive tethered runs have completed without anomaly. Always fly with the safety system armed. Per §3.9, the unified parachute-airbag is the primary passive safety system for all failure modes — loss of lift at any altitude, substrate fracture, water landing, hard impact. The system is built into the central spine and deploys automatically on detected impact or excessive descent rate. Do not disable it for any reason; the CO2 cartridge is inexpensive to replace after a non-event activation. The cymbals can fracture under extreme drive. Sharp bronze fragments at 35 m/s particle velocity are hazardous. Ramp slowly, and inspect for cracks before each flight.

§7 Operating envelope
 Conservatively, the $\chi_v = 0$ baseline hoverbike works: Sea level to ~5 km altitude: easy, ~100 W hover. 5–15 km altitude: workable but power requirement grows with altitude (atmospheric density drops, so more amplitude needed for the same lift). At 15 km: ~300 W. 15–30 km altitude: borderline. Drive amplitudes approach sound speed, entering nonlinear acoustic regime. ~3 kW. Above 30 km: NFAL fails. Atmospheric density too low; substrate amplitudes would have to be supersonic. Vehicle can't hover here without the additional χ_v mechanism (see companion paper 3). At $\chi_v = 0$, this is a ground-effect vehicle. It hovers anywhere the atmosphere has $> \sim 0.02$ kg/m³ density above a reasonably flat surface. That covers all of the troposphere and most

Speed: limited by acoustic-cavity stability during forward motion. The cavity holds shape up to a few m/s lateral velocity; above that, the standing wave starts to leak and lift falls off. This is a cruise-class vehicle, not a fighter jet. Typical operation: 1–5 m/s lateral, 0–30 km vertical envelope.

§8 Conclusion
 The DIY hoverbike is a 150 kg single-rider vehicle that hovers in atmospheric ground effect using nothing but conventional NFAL physics, scaled up via four bronze cymbals and broadband multichord drive. Total bill of materials under \$500. Build time: one weekend. Hover power: ~100–300 W realistic. The same hardware is also the prototype substrate for the broader χ_v experimental program (companion paper 1) and scales up into the saucer-class vehicle that operates outside ground effect (companion paper 3). But the DIY hoverbike as described here doesn't depend on any of that. It hovers on today's physics, using parts from a music store, a hardware store, and a Pimoroni order.

Doctrine sentence · This vehicle works. It hovers 150 kg, in ground effect, on conventional acoustic physics, using parts that anyone can source over a weekend, for under \$500. The role of new physics — if it exists — is to extend the operating envelope above ground effect and beyond the atmosphere. That extension is the subject of companion papers 1 and 3. The vehicle itself, as described here, is buildable today.

References 1. Mueller, B. & Osika, A. (2026a). Probing the OPH Dark Sector via Coherent-Matter Susceptibility. [papers/oph_chi_nu_propulsion_paper.html](#) . (The χ_v physics framework and

experimental measurement program.) **2. Mueller, B. & Osika, A. (2026b). Photonic δ xed-point consensus for SHA-256d proof of work.** Pragma Research / SNRGY Studios. (Where the 12-port substrate adapter the hoverbike reuses was $\delta\delta$ rst speci $\delta\delta$ ed.) **3. Osika, A. (2026c). How to Build a UFO — the Saucer Con δ guration.** papers/how_to_build_a_ufo.html . (The extended architecture across all altitude regimes, space propulsion, and the Tic-Tac-class performance envelope.) patch structure underlying the whole framework.) **5. Brandt, E.H. (2001). “Acoustic physics: Suspended by sound.” Nature 413, 474. (NFAL recap.)** **6. Han, Y. et al. (2020). “Triply periodic minimal surface gyroid structures for additive manufacturing.” Additive Manufacturing 36, 101481.** Draft document. The DIY hoverbike described here is buildable today on conventional physics; nothing in this paper depends on unverified physical hypotheses. The χv measurement scheduled for this afternoon (see companion paper 1) determines whether the same hardware acquires additional capabilities, but the atmospheric hover does not depend on its outcome.